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**PHYSICS**

**9702/21**

Paper 2 AS Level Structured Questions

**May/June 2016**

**MARK SCHEME**

Maximum Mark: 60

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**Published**

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| Page 2 | Mark Scheme                                        | Syllabus | Paper |
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- 1 (a) (i)  $(50 \text{ to } 200) \times 10^{-3} \text{ kg}$  or  $(0.05 \text{ to } 0.2) \text{ kg}$  B1 [1]
- (ii)  $(50 \text{ to } 300) \text{ cm}^3$  B1 [1]
- (b) density = mass/volume or  $\rho = M/V$  C1
- $V = [\pi(0.38 \times 10^{-3})^2 \times 25.0 \times 10^{-2}]/4 (= 2.835 \times 10^{-8} \text{ m}^3)$  C1
- $\rho = (0.225 \times 10^{-3})/2.835 \times 10^{-8}$   
 $= 7940 \text{ (kg m}^{-3}\text{)}$  A1
- $\Delta\rho/\rho = 2(0.01/0.38) + (0.1/25.0) + (0.001/0.225) [= 0.061]$   
or  
 $\%\rho = 5.3\% + 0.40\% + 0.44\% (= 6.1\%)$  C1
- $\Delta\rho = 0.061 \times 7940 = 480 \text{ (kg m}^{-3}\text{)}$
- density =  $(7.9 \pm 0.5) \times 10^3 \text{ kg m}^{-3}$  or  $(7900 \pm 500) \text{ kg m}^{-3}$  A1 [5]
- 2 (a) (i) horizontal component  $(= 12 \cos 50^\circ) = 7.7 \text{ ms}^{-1}$  A1 [1]
- (ii) vertical component  $(= 12 \sin 50^\circ \text{ or } 7.7 \tan 50^\circ) = 9.2 \text{ ms}^{-1}$  A1 [1]
- (b)  $v^2 = u^2 + 2as$  and  $v = 0$  or  $mgh = \frac{1}{2}mv^2$  or  $s = v^2 \sin^2 \theta / 2g$  C1
- $9.2^2 = 2 \times 9.81 \times h$  hence  $h = 4.3 \text{ (4.31) m}$  A1 [2]
- alternative methods using time to maximum height of 0.94 s:
- $s = ut + \frac{1}{2}at^2$  and  $t = 0.94 \text{ (s)}$  (C1)  
 $s = 9.2 \times 0.94 - \frac{1}{2} \times 9.81 \times 0.94^2$  hence  $s = 4.3 \text{ m}$  (A1)
- or  
 $s = vt - \frac{1}{2}at^2$  and  $t = 0.94 \text{ (s)}$  (C1)  
 $s = \frac{1}{2} \times 9.81 \times 0.94^2$  hence  $s = 4.3 \text{ m}$  (A1)
- or  
 $s = \frac{1}{2}(u + v)t$  and  $t = 0.94 \text{ (s)}$  (C1)  
 $s = \frac{1}{2} \times 9.2 \times 0.94$  hence  $s = 4.3 \text{ m}$  (A1)
- (c)  $t (= 9.2/9.81) = 0.94 \text{ (0.938) s}$  C1
- horizontal distance =  $0.938 \times 7.7 (= 7.23 \text{ m})$  C1
- displacement =  $[4.3^2 + 7.23^2]^{1/2}$  C1
- $= 8.4 \text{ m}$  A1 [4]

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- 3 (a) (i) force ( $= mg = 0.15 \times 9.81$ ) = 1.5 (1.47) N A1 [1]
- (ii) resultant force (on ball) is zero so normal contact force = weight  
or  
the forces are in opposite directions so normal contact force = weight  
or  
normal contact force up = weight down A1 [1]
- (b) (i) (resultant) force proportional/equal to rate of change of momentum B1 [1]
- (ii) change in momentum =  $0.15 \times (6.2 + 2.5)$  (= 1.305 N s) C1
- magnitude of force =  $1.305 / 0.12$   
= 11 (10.9) N A1
- or
- (average) acceleration =  $(6.2 + 2.5) / 0.12$  (=  $72.5 \text{ ms}^{-2}$ ) (C1)
- magnitude of force =  $0.15 \times 72.5$   
= 11 (10.9) N (A1)
- (direction of force is) upwards/up B1 [3]
- (iii) there is a change/gain in momentum of the floor M1
- this is equal (and opposite) to the change/loss in momentum of the ball so momentum is conserved A1 [2]
- or
- change of (total) momentum of ball and floor is zero (M1)  
so momentum is conserved (A1)
- or
- (total) momentum of ball and floor before is equal to the (total) momentum of ball and floor after (M1)  
so momentum is conserved (A1)

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- 4 (a) the energy (stored) in a body due to its extension/compression/deformation/  
change in shape/size B1 [1]
- (b) (i) two values of  $F/x$  are calculated which are the same  
e.g.  $10.4/40 = 0.26$  and  $6.5/25 = 0.26$  B1
- or
- ratio of two forces and the ratio of the corresponding two extensions are  
calculated which are the same  
e.g.  $5.2/10.4 = 0.5$  and  $20/40 = 0.5$  (B1)
- or
- gradient of graph line calculated and coordinates of one point on the  
line used with straight line equation  $y = mx + c$  to show  $c = 0$  (B1)
- (so) force is proportional to extension (and so Hooke's law obeyed) B1 [2]
- (b) (ii) 1.  $k = F/x$  or  $k = \text{gradient}$  C1
- gradient or values from a single point used e.g.  $k = 10.4/(40 \times 10^{-2})$
- $k = 26 \text{ N m}^{-1}$  A1 [2]
2. work done = area under graph  
or  $\frac{1}{2}Fx$  or  $\frac{1}{2}(F_2 + F_1)(x_2 - x_1)$   
or  $\frac{1}{2}kx^2$  or  $\frac{1}{2}k(x_2^2 - x_1^2)$  C1
- =  $\frac{1}{2} \times 10.4 \times 0.4 - \frac{1}{2} \times 5.2 \times 0.2$  C1  
or  $\frac{1}{2} \times (5.2 + 10.4) \times 20 \times 10^{-2}$   
or  $\frac{1}{2} \times 26 \times (0.4^2 - 0.2^2)$
- = 1.6 J A1 [3]
- (c) remove the force and the spring goes back to its original length B1 [1]
- 5 (a)  $T = 4 \text{ (ms)}$  or  $4 \times 10^{-3} \text{ (s)}$  C1
- $f = 1/T = 1/0.004$
- = 250 Hz A1 [2]
- (b) intensity  $\propto (\text{amplitude})^2$  and amplitude = 2.8 (2.83) (cm) B1
- curve with same period and with amplitude 2.8 cm B1
- curve shifted 1.0 ms to left or to right of wave X B1 [3]

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- (c) (i) gradient =  $(4.5 - 2.4) \times 10^{-3} / (3.25 - 1.75) [= 1.4 \times 10^{-3}]$  B1
- wavelength =  $0.45 \times 10^{-3} \times 1.4 \times 10^{-3}$  C1
- =  $6.30 \times 10^{-7}$  (m) C1
- = 630 nm A1 [4]
- (ii) (gradient is equal to  $\lambda/a$  therefore) gradient of line is reduced B1
- value of  $x$  will be reduced for all values of  $D$   
or new line is completely below old line  
or intercept is less B1 [2]
- 6 (a) (coulomb is) ampere second B1 [1]
- (b) (total) charge or  $Q = nAle$  M1
- $I = Q/t$  and  $l/t = v$  M1
- $I = nAle/t = nAve$  therefore  $v = I/nAe$  A1 [3]
- (c) (i) ratio =  $(I/nA_Y e) / (I/nA_Z e)$  C1
- =  $A_Z/A_Y$  or  $4A/A$  or  $\pi d^2 / (\pi d^2 / 4)$  C1
- = 4 A1 [3]
- (ii)  $R = \rho l/A$  or  $R = 4\rho l / \pi d^2$  B1
- $R_Y = \rho l/A$  and  $R_Z = \rho(2l)/4A$  so  $R_Y/R_Z = 2$   
or  
 $R_Y = 4\rho l / \pi d^2$  and  $R_Z = 4\rho(2l) / \pi 4d^2$  or  $2\rho l / \pi d^2$  so  $R_Y/R_Z = 2$  A1 [2]
- (iii)  $V = 12R_Y / (R_Y + R_Z)$  or  $I = 12 / (R_Y + R_Z)$  and  $V = IR_Y$  C1
- $V = 12 \times 2/3$   
 $= 8(0)V$  A1 [2]
- (iv) ratio =  $I^2 R_Y / I^2 R_Z$  or  $(V_Y^2 / R_Y) / (V_Z^2 / R_Z)$  or  $(V_Y I) / (V_Z I)$
- = 2 A1 [1]

| Page 6 | Mark Scheme                                        | Syllabus | Paper |
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- 7 (a) hadron: neutron/proton  
and  
lepton: electron/(electron) neutrino  
*(allow other correct particles)* B1 [1]
- (b) (i) proton: up up down or uud B1 [1]  
(ii) neutron: up down down or udd B1 [1]
- (c) (i) neutron  $\rightarrow$  proton + electron + (electron) antineutrino B1 [1]  
(ii) up down down (quarks) change to up up down (quarks)  
or  
down (quark) changes to up (quark) B1 [1]